

Number Theory

Reflection

This was more like an extension to the number theory related units that I learned in SIT192 – Discrete Maths

Response to Feedback

To show that each learning objective has been achieved, please divide both the summary and the evidence of learning into sections aligned with the learning objectives. You can find the template in Task List > Number Theory > Resources.

Done. I have changed the order and updated the titles accordingly. The content itself has not been changed.

Basics

Congruence

When $a, r, m \in \mathbb{Z}$, and $m > 0$, and if m divides $(a - r)$ we can say the statement below. This basically means that a and r leave the same remainder when divided by m .

$$a \equiv r \pmod{m} \text{ when } m \mid (a - r)$$

To reduce a number modulo m , just divide it by m and keep the remainder. For example,

$$-27 \equiv 34 \pmod{61}, \text{ because } 34 - (-27) = 61.$$

Euclidean algorithm

$\gcd(a, b)$ is the largest positive integer that divides both a and b . Two numbers are said to be relatively prime when their $\gcd$ is 1. This check is important for modular inverse, Fermat's Little Theorem, Euler's Theorem and the Chinese Remainder Theorem down the line.

When $b > a$, do this until $r_{i+1} = 0$. Then, r_i is the $\gcd$.

$$\begin{aligned} \text{rem}(b \div a) &= r_1 \\ \text{rem}(a \div r_1) &= r_2 \\ \text{rem}(r_1 \div r_2) &= r_3 \end{aligned}$$

A handwritten diagram illustrating the steps of the Euclidean algorithm. It shows three equations stacked vertically: $b - a \times m = r_1$, $a - r_1 \times m_1 = r_2$, and $r_1 - r_2 \times m_2 = r_3$. Red arrows indicate the substitution of a from the first equation into the second, and r_1 from the second equation into the third. Green arrows indicate the substitution of r_1 from the first equation into the second, and r_2 from the second equation into the third.

Obtain the Bezout's coefficients of a pair of numbers.

If $\gcd(a, b) = c$, then, $m \cdot a + n \cdot b = c$

a and b are relatively prime only if numbers m and n suffice $m \cdot a + n \cdot b = 1$

Instead of trying to guess m and n by substituting a and b, just re-arrange the equations. Use the $b - a \cdot m = r_1$ method (Euclidian algorithm), and substitute values from where necessary and keep on doing that until the final equation comes to the form said above, with m and n being real integers.

Apply the extended Euclidean algorithm for computing the modular inverses.

The modular inverse of a modulo m is a number x such that: $a \cdot x \equiv 1 \pmod{m}$. A modular inverse exists if and only if $\gcd(a, m) = 1$ (aka. they are relatively prime).

$$\begin{aligned} p &\equiv m \text{ has an inverse } \pmod{p} \\ q &\equiv \gcd(m, p) = 1 \\ p &\leftrightarrow q \\ \text{By Bezouts'} &\Rightarrow \\ \text{if the } \gcd(m, p) &= 1, \\ \text{there are 2x numbers 'a' and 'b':} \\ am + bp &= 1 \\ \text{we use extended euclidean algorithm} \\ am - 1 &= -bp \\ \hookrightarrow a \text{ is a multiple of } p \\ \hookrightarrow am &\equiv 1 \pmod{p} \\ &\quad \uparrow \\ &\quad \text{inverse} \end{aligned}$$

To find the inverse of a **89 mod 28**, find m and n (Bezouts coefficients) such that **89m + 28n = 1**. So in this case, it would be **1 = -11*89 + 35*28**. This means that -11 is an inverse of **89 mod 28**. Also note that finding just the Bezouts coefficient of **89** (-11 in this case) is enough just for the sake of solving these quickly. As -11 is out of the 0-27 range (for mod 28), we will reduce it to get **-11+28 = 17**. Now, you can say that the inverse of 89 is **17 (mod 28)**.

Solving Linear Equations in Modular Arithmetic

If $\gcd(m, p) = 1$, then for any number x the equation $mx = a \pmod{p}$ has a solution.

To solve $a \cdot x \equiv b \pmod{m}$, first check $\gcd(a, b)$. If this is equal to 1, find the inverse of **a** and multiply both sides by it. For example, let's look at **70x ≡ 41 (mod 61)**:

- Reduce 70 mod 61 $\rightarrow 70 \equiv 9 \pmod{61}$
- Use the extended euclidean algorithm $\rightarrow 1 = 4 \cdot 61 - 27 \cdot 9$
- The inverse of 9 is $-27 \equiv 24 \pmod{61}$
- Multiply both sides by 34: $x \equiv 34 \cdot 41 = 1394 \equiv 52 \pmod{61}$.

Or when generalized, it will look as follows, to solve $ax \equiv b \pmod{p}$ if $\gcd(a, p) = 1$,

- Find the inverse s of a modulo p using the extended Euclidean algorithm.
- Multiply both sides of the equation by s : $asx \equiv bs \pmod{p}$.
- This results in the solution $x \equiv bs \pmod{p}$.

Applying the Chinese remainder theorem to solve systems of two linear congruences.

This theorem lets us combine two modular equations into one equation, as long as the two moduli have no common factor other than 1. In other words, the gcd of the two moduli should be 1 to apply this algorithm.

For the two equations:

$$x \equiv a \pmod{m}$$

$$x \equiv b \pmod{n}$$

First, check if $\gcd(m, n) = 1$. If it is, proceed. Then, use the extended Euclidean algorithm to find $m(s)$ and $n(t)$.

$$m*s + n*t = 1$$

Using these values, substitute them to the final formula to get the solution.

$$x \equiv bms + ant \pmod{mn}$$

When there are three equations, solve the first two and then solve it again using its solution and the third equation.

Fermat's Little Theorem

When p is prime and p does NOT divide a , Fermat's Little Theorem says that:

$$a^{p-1} \equiv 1 \pmod{p}$$

This lets us use a large exponent modulo $p-1$. For example, let's look at **$11^{27699} \bmod 13$** .

- $11^{12} \equiv 1 \pmod{13}$, because 13 is prime
- And, $27699 = 12 * 2308 + 3$
- So,
 - o $11^{27699} \equiv (11^{12})^{2308+3}$
 - o $11^{27699} \equiv (11^{12})^{2308} \times 11^3$
 - o $11^{27699} \equiv 11^3$
 - o $11^{27699} \equiv 5 \pmod{13}$

However, if the p is not a prime number, you will have to use another method using the Euler's Totient Function. Only the first part changes, the rest of it remains almost identically.

Proof of $a^p \equiv a \pmod{p} \rightarrow$

$a^p \equiv a \pmod{p}$ — p is prime, a is $\times$ a multiple of p

1. $k \cdot a = r_k \pmod{p}$ — for every k from 1 to $p-1$

2. we don't get $r_k = 0$,

because $(k \cdot a - r_k)$ is a multiple of p

$k \cdot a$ is not a multiple of p (as p is prime)

$k \cdot a$ is " " " " (as it's smaller than p)

why $r_k \neq r_j$ if $k \neq j$?

$$[(k-j) \cdot a - (r_k - r_j)]$$

not a multiple of p

r_k 's take all values between 1 & $p-1$

3. multiply on both sides

$$\prod_{k=1}^{p-1} k \cdot a \equiv \prod_{k=1}^{p-1} r_k \pmod{p}$$

$$a^{p-1} \prod_{k=1}^{p-1} k \equiv \prod_{k=1}^{p-1} k \pmod{p}$$

$$a^{p-1} \equiv 1 \pmod{p}$$

$$a^p \equiv a \pmod{p}$$

Euler's Totient Function: ϕ

$\phi(n)$ counts the integers from 1 to $n-1$ that are relatively prime to n . To calculate this, either list out all the numbers from 1 to $n-1$ (excluding n and 0) and count the relatively prime numbers to n manually.

Or, you can break n down into its prime factors (powers), and use this algorithm.

$$\text{If } n = p_1^{a_1} \times p_2^{a_2} \times \dots \times p_k^{a_k}$$

$$\text{Then, } \phi(n) = n \cdot \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

Perform prime factorization for n , take each prime factor p (without the powers) and multiply n by $\left(1 - \frac{1}{p}\right)$.

Euler's Theorem

When $\gcd(a, n) = 1$, Euler's theorem says:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Fermat's Little Theorem is the prime version of Euler's Theorem. This is because $\phi(p) = p-1$ when p is prime. To simplify a large power, divide the exponent by $\phi(n)$ and keep the remainder

Examples

Fermat's Little Theorem & Euler's Totient Function

3. $7^{31317} \bmod 10$ — not prime

$$\gcd(7, 10) \Rightarrow 10 - 7 \times 1 = 3 \Rightarrow 1$$

$$7 - 3 \times 2 = 1$$

$$\phi \Rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$\uparrow \quad \uparrow \quad \quad \uparrow \quad \quad \uparrow$

$$\therefore \phi(10) = 4$$

$$\text{Euler's Theorem} \Rightarrow 7^4 \equiv 1 \pmod{10} \quad \left| \begin{array}{l} 31317 = 4 \times 7829 + 1 \end{array} \right.$$

$$\begin{aligned} 7^{31317} &= 7^{4(7829) + 1} \\ &= (7^4)^{7829} \times 7 \\ &\equiv 1^{7829} \cdot 7 \pmod{10} \\ &\equiv 7 \pmod{10} // \end{aligned}$$

Exercise 1.4.4. Calculate:

1. $11^{27699} \bmod 13$ — prime

$$a^p \equiv a \pmod{p}$$

$$\gcd(11, 13) \Rightarrow 13 - 11 \times 1 = 2 \Rightarrow 1$$

$$11 - 2 \times 5 = 1$$

$$11^{13-1} \equiv 1 \pmod{13}$$

$$11^{12} \equiv 1 \pmod{13}$$

$$\left| \begin{array}{l} 27699 - 12 \times 2308 = 3 \\ 27699 = 12 \times 2308 + 3 \end{array} \right.$$

$$\begin{aligned} \therefore 11^{27699} &= 11^{12 \cdot 2308 + 3} \\ &= (11^{12})^{2308} \times 11^3 \\ &\equiv 1^{2308} \cdot 11^3 \pmod{13} \\ &\equiv 11^3 \pmod{13} \\ &\equiv 5 \pmod{13} // \end{aligned}$$

$$\left| \begin{array}{l} 11^3 \equiv (-2)^3 \\ \equiv -8 \\ \equiv 5 \pmod{13} \end{array} \right.$$

Finding the Last Digits of a Large Power

For the last k digits of a number n , calculate: $n \bmod 10^k$.

- For the last digit: **mod 10**.
- For the last two digit: **mod 100**.
- For the last three digit: **mod 1000**.